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PDF

introduction, problem statement, objective, motivation,
base paper, scope, novelty, POs, PSOs, SDG Mappint,
references

Below is a **project-ready write-up** for your topic:

AI Agent–Enabled Smart Water Level and Leakage Detection System for Residential Areas in Mysuru

I have structured it exactly in the format you asked: **Introduction, Problem Statement, Objectives, Motivation, Base Paper, Scope, Novelty, POs, PSOs, SDG Mapping, and References.**

1. Introduction

Water scarcity and water wastage are major issues in rapidly growing urban areas such as Mysuru. Many residential buildings face problems like water overflow from tanks, undetected pipeline leakages, and inefficient water usage monitoring. Traditional monitoring systems depend on manual observation, which often leads to delays and water loss.

With advancements in **Artificial Intelligence (AI), Internet of Things (IoT), and smart sensors**, it is possible to automate water management. An AI-enabled system can continuously monitor water level and detect leakage patterns in real time, allowing early alerts and intelligent decision-making. The proposed system integrates sensors, AI-based analysis, and real-time notifications to ensure efficient water usage and reduce wastage.

2. Problem Statement

Residential areas in Mysuru experience:

- Water wastage due to tank overflow
- Hidden leakages in pipelines causing loss of water

- Lack of real-time monitoring systems
- Delayed manual detection and maintenance
- Increasing demand for sustainable water usage

Hence, there is a need for an **intelligent, real-time automated water monitoring system** capable of detecting abnormal usage and leakage conditions using AI techniques.

3. Objectives

The main objectives are:

1. To design an IoT-based system for monitoring water tank levels continuously.
 2. To detect leakage or abnormal water flow using AI-based analysis.
 3. To provide real-time alerts through mobile or web applications.
 4. To reduce water wastage and improve water management efficiency.
 5. To create a scalable solution suitable for residential areas in Mysuru.
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4. Motivation

The motivation behind this project includes:

- Increasing water scarcity and urban consumption.
- Need for sustainable smart-city solutions.
- Frequent water overflow and leakage problems in apartments and houses.
- Availability of low-cost IoT sensors and AI algorithms.
- Support for SDG goals related to water conservation.

Additionally, modern AI systems can analyze sensor data patterns similarly to how AI models analyze sequential data in other domains (e.g., real-time detection systems). The idea of combining feature extraction with intelligent models is inspired by recent AI research approaches that emphasize real-time prediction and contextual analysis.

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5. Base Paper

Base Paper Title:

A voice-based real-time emotion detection technique using recurrent neural network empowered feature modelling

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Why this paper is used as base:

Although the paper focuses on emotion detection, it provides important concepts relevant to your project:

- Real-time monitoring and prediction techniques
- AI-based feature extraction from continuous data streams
- Use of sequential models for pattern detection
- Context-aware prediction mechanisms

Key Idea Adapted:

- The paper uses **feature extraction + sequential model analysis** for real-time decision-making.
 - Similarly, your project will use sensor data features (water level, flow rate, pressure) and AI models to detect anomalies like leakages and overflow in real time.
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6. Scope of the Project

The project scope includes:

- Smart water tank level monitoring
- Pipeline leakage detection using sensors
- AI-based anomaly detection
- Notification system for residents
- Data logging for usage analysis

Future Scope

- Integration with smart city dashboards
 - Predictive water demand forecasting
 - Automatic motor ON/OFF control using AI
 - Integration with municipal water management systems
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7. Novelty of the Proposed System

The novelty of this project includes:

- Combination of AI agents with IoT-based water monitoring
- Real-time leak detection instead of manual inspection
- Intelligent pattern recognition rather than simple threshold detection
- Low-cost and scalable implementation for residential areas
- Smart alerts and automation capability

Unlike traditional systems that only show water level, this system learns patterns and detects unusual behaviour intelligently.

8. Program Outcomes (POs) Mapping

PO	Description	Contribution
PO1	Engineering knowledge	Application of IoT & AI concepts
PO2	Problem analysis	Identifying leakage and overflow issues
PO3	Design solutions	Developing smart monitoring system
PO4	Investigation	Data collection and testing
PO5	Modern tool usage	Sensors, cloud, AI tools
PO6	Society impact	Water conservation
PO7	Environment sustainability	Reducing water wastage
PO9	Teamwork	Hardware + software integration
PO10	Communication	Alert and reporting system
PO12	Lifelong learning	Emerging smart technologies

9. Program Specific Outcomes (PSOs) Mapping

PSO	Description	Mapping
PSO1	Apply core engineering concepts	Sensor integration and data processing
PSO2	Design intelligent systems	AI-enabled anomaly detection
PSO3	Develop practical solutions	Real-time monitoring application

10. SDG (Sustainable Development Goals) Mapping

SDG	Relevance
SDG 6 – Clean Water and Sanitation	Efficient water usage and leakage prevention
SDG 11 – Sustainable Cities and Communities	Smart residential infrastructure
SDG 12 – Responsible Consumption and Production	Reducing resource wastage
SDG 13 – Climate Action	Sustainable water management

11. References

1. Chamishka, S., et al., "A voice-based real-time emotion detection technique using recurrent neural network empowered feature modelling," *Multimedia Tools and Applications*, 2022.  97aac2fc-1e38-4203-8602-5acc3ee...
2. IoT-based smart water monitoring research papers (IEEE/Elsevier).
3. Smart city water management literature.
4. AI and anomaly detection system studies.

★ If you want, I can ALSO give you:

- 🔥 VTU project report ready format (Chapter-wise)
- 🔥 Block diagram + Architecture diagram for your system
- 🔥 Problem-to-solution flowchart
- 🔥 Methodology section (easy 8–10 marks answer style)
- 🔥 Literature survey table (very useful for project reviews)

Just tell me 👍.

Sources



A voice-based real-time e...
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Here is a **clean, project-report ready version** of your content (formatted professionally for VTU / final year project documentation).

AI Agent–Enabled Smart Water Level and Leakage Detection System for Residential Areas in Mysuru

1. Introduction

Water management has become a critical challenge in urban residential areas due to increasing population and growing demand. In cities like Mysuru, water wastage caused by tank overflow, leakage in pipelines, and lack of monitoring systems leads to significant resource loss. Traditional systems depend on manual checking, which is inefficient and error-prone.

Recent advancements in **Artificial Intelligence (AI)**, **Internet of Things (IoT)**, and **real-time monitoring systems** enable smart solutions that can continuously observe sensor data, detect abnormal patterns, and take proactive actions. Similar to AI-based real-time prediction models used in continuous data analysis, intelligent systems can extract features and make contextual decisions from live data streams.  A voice-based real-time emotion...

This project proposes an AI-enabled system that monitors water level and identifies leakage in real time to ensure efficient water usage in residential environments.

2. Problem Statement

Residential areas in Mysuru face several water-related problems such as:

- Overflow of overhead tanks leading

Sources



I've given you a new research paper, give me details of that paper

Here are the **complete details of the new research paper** you provided, explained in a clear and student-friendly way (useful for project review, literature survey, and viva).

Research Paper Details

Paper Title

"A voice-based real-time emotion detection technique using recurrent neural network empowered feature modelling"  A voice-based real-time emotion...

1. Basic Information

- **Authors:**
 - Sadil Chamishka
 - Ishara Madhavi
 - Rashmika Nawaratne
 - Damminda Alahakoon
 - Daswin De Silva
 - Naveen Chilamkurti
 - Vishaka Nanayakkara
 - **Published In:** *Multimedia Tools and Applications* (2022)  A voice-based real-time emotion...
 - **Institutions:**
 - University of Moratuwa
 - La Trobe University
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2. Introduction (Paper Idea)

The paper focuses on detecting **human emotions from voice conversations** using AI. With the growth of IoT and multimedia systems, huge volumes of voice data are generated daily. Detecting emotions from speech helps machines better understand humans and improve interaction quality.  A voice-based real-time emotion...

The authors highlight that:

- Text-based emotion detection performs better than audio-only methods.
- Audio-based systems still suffer from lower accuracy.

- Real-time emotion prediction is important for applications like call centers and healthcare.

3. Problem Addressed

The paper identifies these key issues:

- Limited number of emotions detected in existing systems.
- Low performance of audio-based emotion detection compared to text.
- Need for speaker separation (speaker diarization).
- Lack of contextual understanding in conversation flow.  A voice-based real-time emotion...

4. Objective of the Paper

Main goals include:

1. Improve emotion detection accuracy using audio features.
2. Introduce a new feature representation called **Bag-of-Audio-Words (BoAW)**.
3. Use a context-aware **Recurrent Neural Network (RNN)** model.
4. Enable **real-time emotion recognition**.

5. Proposed Methodology

The system works in **three major phases**:

(a) Feature Extraction

- Uses openSMILE toolkit.
- Extracts 130 low-level audio features every 10 ms.
- Features include:
 - MFCC
 - Pitch
 - Energy
 - Spectral information
 - Voice quality features  A voice-based real-time emotion...

≡ ChatGPT 5.2 ▾

↑ ...

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- Inspired by bag-of-words from NLP.
 - Converts variable-length audio into fixed-size feature vectors.

- Creates a codebook of frequent audio patterns.
- Improves representation quality and reduces noise.

(c) RNN-Based Emotion Detection

- Uses **BiDialogRNN** with attention mechanism.
 - Captures:
 - Global conversation context
 - Speaker state
 - Emotion history
 - Predicts emotions using sequence learning.
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6. Emotion Categories Detected

The model predicts six basic emotions:

- Happy
 - Sad
 - Neutral
 - Angry
 - Excited
 - Frustrated  A voice-based real-time emotion...
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7. Key Technical Contributions

The paper contributes:

1. New **BoAW feature embedding** for speech.
 2. Improved **attention mechanism** aligned with feature representation.
 3. Context-aware emotion prediction using conversation history.
 4. Real-time prediction capability validation.  A voice-based real-time emotion...
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8. Datasets Used

IEMOCAP Dataset

- Dyadic conversations.
- Emotion-labelled audio samples.
- Used for primary evaluation.

2 MELD Dataset

- Multi-speaker conversations.
 - More challenging because of rapid emotion changes.  A voice-based real-time emotion...
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9. Results

The proposed model achieved:

- **Weighted Accuracy:** 60.87%
- **Unweighted Accuracy:** 60.97%

These results outperform previous audio-only emotion detection models.

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The paper also shows:

- Better performance using BoAW than baseline systems.
- Improved context understanding via attention mechanisms.
- Real-time latency $\approx$ **0.7 seconds per utterance**, suitable for live applications.

 A voice-based real-time emotion...

10. Applications Mentioned

- Automated customer call centers
 - Healthcare monitoring
 - Human-machine interaction
 - Elderly care systems
 - AI conversational agents
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11. Limitations

The authors mention:

- Need for better real-time speaker diarization.
 - Accuracy still lower than multimodal (audio + text + video) systems.
 - Dataset imbalance affects results.
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12. Future Work Suggested

- Mixed emotion detection.
 - Integration with text features.
 - Use of advanced speech models like wav2vec.
 - Larger audio datasets for training.  A voice-based real-time emotion...
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13. Key Takeaway (Important for Your Project)

Although this paper focuses on emotion detection, the **core idea is very relevant** to your smart water project:

- Real-time data monitoring
- Feature extraction from continuous signals
- AI-based pattern learning
- Context-aware prediction

You can use the same concept for:

- Detecting water leakage patterns
 - Recognizing abnormal flow behaviour
 - Smart decision-making using AI
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★ If you want, I can also give you:

- 🔥 Literature survey paragraph (VTU format) based on this paper
- 🔥 How this paper connects to your Smart Water project (exact mapping)
- 🔥 Base paper analysis table (very important for project guide reviews)
- 🔥 5-minute viva explanation of this paper in simple words

Just tell me 👍.

Sources

